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PALEOMAGNETISM AND MAGNETOSTRICTION

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ABSTRACT

The rôle of magnetostriction in the problem of rock magnetism has heretofore been neglected. Based on some experimental observations and on reconsiderations of well-known factors affecting rocks, the conclusion is reached that magnetostriction may figure prominently in establishing the directions of magnetization observed in many cases. Therefore, the practice of interpreting meager magnetic data in terms of polar wandering and continental drift can be in error.

During the course of a visit last winter with Prof. Hales and his colleagues at the Bernard Price Institute for Geophysical Research in South Africa, the writer became concerned about the problem of grossly scattered magnetization directions noted in various surface samples of the Karroo dolerites. The usual intuitive feeling is that igneous rocks which receive their primary magnetizations at the time of their cooling should be uniformly magnetized, but actually the inconsistent magnetizations observed were strongly reminiscent of the distributions noted in some conglomerates. It was reasonable, therefore, to spend some time groping for plausible mechanisms by which scattered directions of magnetization might be explained. The result of these efforts is the realization that magnetostriction may be a factor of great importance in paleomagnetic studies.

Consider the thermal and mechanical history of a thin horizontal plate of dolerite which is cooling after having been injected molten into the inside of a thick sequence of bedded sediments. Within the melt at high temperature, hydrostatic equilibrium prevails. During the cooling and solidification process, the overburden of sediments encourages the intrusive to contract in the vertical direction, but there is no corresponding way in which it can accommodate contraction in the horizontal plane. As a consequence, cracks develop which divide the now-solidified plate into myriads of vertical columns. These columns now are supporting the overburden of sediments, and hence they are in axial compression. At the completion of cooling, we may imagine that the dolerite sill is magnetized uniformly in the direction of the earth's magnetic field, in spite of the fact that the axial compression will lower the symmetry of cubic ferromagnetic minerals and perhaps impose a preferred orientation of exsolution lamellae of ülvospinel, ilmenite, etc. With the passage of time and erosion processes, the sill is unburdened, and ultimately a sample is removed for measurement. The sample is now no longer under axial compression and is free to recover from elastic strains. Depending on

its magnetostrictive properties, the unloading may cause a new magnetic moment to be added to the original one, and the observed magnetization would then have no simple and obvious relationship to the magnetic field that was applied during the cooling. It is not yet known whether such a process accounts, for example, for the nearly vertical polarizations observed in Tasmanian dolerites [see 1 of "References" at end of paper].

To investigate the question of whether rocks might be significantly magnetostrictive, some exploratory experiments were carried out at the Department of Terrestrial Magnetism with a simple astatic magnetometer and a dural framework with which axial compression could be applied to 1″ by 1″ cylindrical samples by means of a push-rod having at its remote end a hydraulic jack. Although this equipment was adequate in that it caused no magnetometer deflections by itself, it was not refined to the point where precise measurements of directions and intensities were considered possible. Nevertheless, it was useful in showing the following:

(1) That 3,000 pounds/inch^2 axial compression is adequate to impart to many samples a new magnetic moment which is comparable to the natural moment of the sample; and
(2) The new moment added is proportional to the applied compression, and disappears when the sample is released (in short-time experiments).

The samples tested were of various basic igneous rocks and had magnetic moments greater than 10^{-5} cgs/cc. That there was considerable variation in their magnetostrictive behavior is no surprise in view of what is known about the dependence of magnetostriction constants on composition in ferrites.

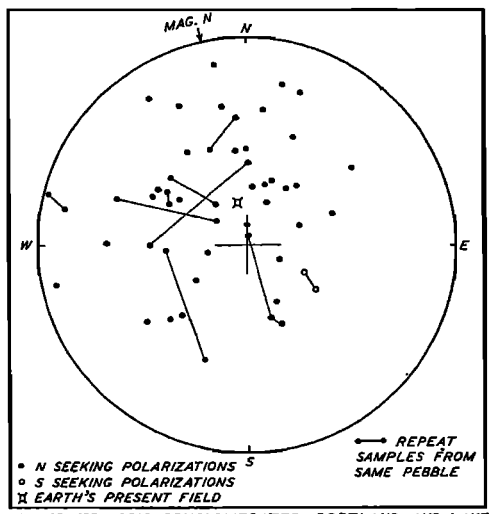
Returning now to the scattered magnetizations of the dolerites, the fact noted and stressed by Gough that surface samples give scattered results and samples from depth in the same structure give consistent results has been amplified by Hales and Nairn. Weathering and lightning have generally been held responsible for this effect, but the mechanism by which the former might change the magnetic moment has never been clarified. Apart from color changes, the most conspicuous visible effect of weathering of basic igneous rocks in dry climates is in the hydration and alteration of feldspars and pyroxenes. The hydrated rock has a lower density than its parent, and because of size considerations it characteristically spalls off as sheets along curved surfaces roughly parallel to the exposed surface. Obviously, large forces have acted when a rock sheds an outer layer of tough material a few inches thick. Because of the steep stress gradient near the surface of a weathering rock, the usual sample prepared for measurement may retain stresses. These, it would seem, are adequate by magnetostriction to cause haphazard deflections of the otherwise uniform magnetization of the rock. Thermal cycling by the heat of the sun may help the rock to maintain a magnetic vector which has been deflected by magnetostriction.

It is appropriate to wonder whether the scattered magnetizations observed in some conglomerates should be explained on the basis of magnetostriction. If the conglomerate consists mainly of large fragments with little cementing matrix, it is certain that the stresses acting on any chosen piece are distinctly non-hydrostatic. Situations can be visualized where the magnetizations are quite

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unstable and over a period of time come slowly into equilibrium with the earth's field while the fragments are non-uniformly stressed. If then these fragments are removed relatively quickly from their surroundings, by magnetostriction, they can acquire new directions of magnetization which are not parallel to the applied field. Following the usual interpretation, such samples would be said to have some magnetic stability, whereas actually all the fragments were, in this example, magnetized parallel to one another and the earth's field while *in situ*. Measurements made on different specimens from the same cobble would not be expected to agree with one another because of the non-uniform stresses.

Such a situation may have been encountered in some Triassic conglomerates in the eastern United States (Fig. 1). Repeat samples from the same cobble show



UPPER TRIASSIC CONGLOMERATES, PORTLAND AND LAKE QUONNIPAUG, CONN.; SUFFERN, N.Y.

FIG. 1

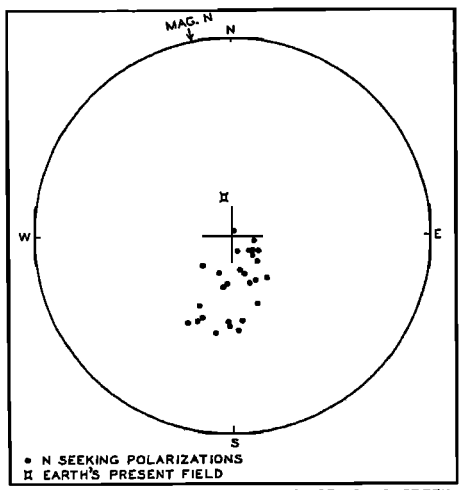
considerable differences in direction, and, furthermore, the distribution of directions does not follow the scatter pattern that would be expected if initially the magnetizations had started at random and had turned toward the earth's field at rates proportional to the sine of their angular departures from the field (magnets in tar model).

The demonstration that the magnetic moments of rocks may be significantly pressure-dependent calls for imaginative and critical consideration. The generalization may be ventured that there are no old rigid (that is, non-plastic) rocks exposed on the surface of the earth which have not been broken more or less systematically by joint planes into fairly small pieces (of the order of tens of feet or smaller). The very fact that these rocks have been broken implies that they have been subjected to large non-hydrostatic forces. The influence of these forces on the final magnetic moment we measure must be considered. At this time, we have no assurance that the long-term action of stress on the magnetic moment is completely reversible and that its final effect is nil when the stress

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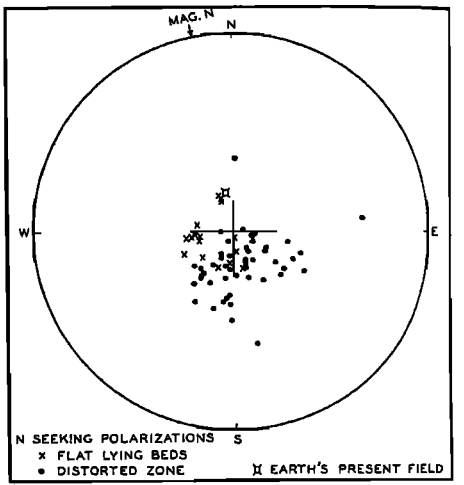
is released. It seems prudent to anticipate that any geologic forces which were capable of breaking rocks were likely able to produce permanent changes in their previous magnetizations.

Possibly as a case in point, we may consider the magnetizations observed in some flat-lying limestones in the eastern United States in New York State. Samples taken in a Trenton limestone conglomerate (Ordovician age) show polarizations dipping steeply toward the south in a reasonable cluster (Fig. 2). Approximately 50 miles to the east, the Onondaga limestone (Devonian) in two different settings



TRENTON LIMESTONE CONGLOMERATE, FLAT CREEK, SPRAKERS, N.Y., JULY 1954

FIG. 2



POLARIZATIONS OF FLAT LYING AND DISTORTED ONONDAGA LIMESTONE, NEW SALEM AND CLARKSVILLE N.Y., JULY 1954

FIG. 3

has the same pattern of magnetization in about the same orientation (Fig. 3). Except that all these samples are uniformly high in acid soluble carbonates, they are quite variable, particularly in color and insoluble residues. Post-depositional magnetization is indicated by the observations in the conglomerates and in the distorted zone (penecontemporaneous deformation indicated by distorted bedding containing undeformed bivalve shells and corals). That the Devonian beds have never been heated appreciably has been demonstrated on the basis of the special assortment of amino acids they contain at a concentration of only a few parts per million. The possibility that the clustering of observed directions was an artifact resulting from a systematic bias in the sampling or measuring techniques was eliminated by proper sampling procedures.

These magnetizations, according to the currently popular methods of rock magnetism, might be interpreted in one or several of the following ways:

(1) At some time since the lower Devonian, the earth's magnetic dipole axis emerged in the vicinity of the Caribbean Sea with a sufficient intensity for a long enough time that the rocks acquired, even though cool, an approximately uniform isothermal remanent magnetization which remained stable in subsequent time.

(2) The steep polarizations toward the south represent a vector sum of magnet-

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izations imposed by the present earth's field and magnetizations imposed at some other time when the dipole axis emerged somewhere to the morth or south of the outcrop. The various samples are sufficiently identical magnetically that in all of them the two vectors have maintained about the same ratio.

(3) This section of New York State was formerly on the opposite side of the geographic pole and has been translated along a meridian some 90°.

(4) This section of New York State has been rotated 180° on a vertical axis.

(5) Combinations of these processes have affected these rocks in unknown proportions and they are therefore not suitable for paleomagnetic purposes.

Serious specific objections can be leveled against the first four interpretations, and the value of the fifth cannot be judged at this time. In summary, the findings in these rocks are in gross conflict with currently held views (magnetostriction excepted) or common sense, or both. It seems that something must be added to our knowledge of rock magnetism.

It cannot be claimed at this time that in magnetostriction the full explanation of these puzzles has been found. It seems, however, that it may provide a better basis for understanding discrepancies and complications in meager magnetic data than the currently popular practice of interpreting magnetic evidence as support of the hypotheses of continental drift and polar wandering.

It is proposed here, for the sake of argument, that the pattern of magnetization displayed by these rocks is in some way related to the unloading and shear, and to the uniform deformation that has produced in them simple systems of vertical joints. This deformation was not sufficiently severe to produce observable folds or important faults, but was nevertheless adequate to reduce the once continuous strata into a mosaic pattern of blocks. In places, the spacing of these joints is only a matter of inches, and only rarely can one find an unbroken piece as long as 25 feet. The questions of when and how and under what conditions these joints were produced lead to many complex considerations (see, for example, reference [2]), but the very fact that the rocks are jointed indicates that they have been subjected to large stresses. The fact that in short-time laboratory experiments magnetostriction is reversible is no basis for concluding that the effect of long applied stresses will vanish when the sample is unloaded.

The writer's present feeling is that, although many of the available rock magnetism data are consistent with various (unrelated) versions of the hypotheses of continental drift and polar wandering, they by no means prove that such important processes have taken place. The data are tantalizing, but they must be greatly amplified and strengthened. We must look for wide regional consistencies in the magnetization patterns of rocks of varied types in varied settings, and we must gain accurate and ample knowledge of the processes by which rocks are magnetized.

References

[1] E. Irving, The magnetization of the Mesozoic dolerites of Tasmania, Proc. R. Soc. Tasmania, **90**, 157-168 (1956).

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